Harmonizing gsm and the Pharmaverse: Orchestrating SDTM, ADaM, TFLs, and ARS in a Unified R Workflow

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Background

The `pharmaverse` has rapidly emerged as an open-source standard for clinical trial reporting. Core packages such as `sdtm.oak` support SDTM transformations, while `admiral` provides a robust framework for ADaM derivations. Downstream, a rich ecosystem of visualization tools enables production-ready outputs — including tabular summaries with `gtsummary` and safety visualizations via `safetyCharts`. At the same time, the CDISC Analysis Results Standard (ARS) is gaining momentum, with packages like `cards` beginning to support ARS-aligned workflows. Complementing traditional static deliverables, many organizations are increasingly adopting Shiny and web-based review tools to enable interactive exploration of clinical data.

In parallel, the gsm framework — particularly the gsm.core package — has matured into a YAML-driven workflow engine designed to deliver reproducible, traceable, and audit-ready pipelines for clinical data science. The framework currently powers a standardized Risk-Based Quality Monitoring (RBQM) infrastructure, pairing flexible data workflows with robust reporting systems such as those demonstrated in `gsm.kri`.

Historically, the `pharmaverse` and `gsm` ecosystems have evolved along separate trajectories. However, their complementary strengths present a compelling opportunity for integration: combining pharmaverse's domain-specific data standards and reporting capabilities with gsm's workflow orchestration and auditability could form the foundation of a fully traceable, end-to-end clinical reporting pipeline.

With the introduction of {workr} as a standalone workflow engine, we can now explore how these ecosystems can harmonize to deliver cohesive, reproducible pipelines that lower barriers to adopting open-source R infrastructure in regulated clinical trial reporting.

Objective

We present a pipeline that demonstrates how {workr} can orchestrate `pharmaverse` packages across the full reporting lifecycle — transforming raw eCRF data into SDTM, ADaM, TFL outputs, and ARS-aligned deliverables. Through this case study, we illustrate how harmonizing these ecosystems enables a cohesive, reproducible workflow that lowers barriers to adopting open-source R infrastructure in regulated clinical trial reporting.

Methods

- {workr} engine: In this framework, workflows are defined declaratively using YAML configuration files organized into three sections:

1. Meta — captures descriptive metadata and documentation associated with the workflow

2. Spec — declares required data sources and their structural requirements, defining the inputs needed for downstream derivations and transformations
3. Steps — specifies the execution pipeline. Each step is written as an {output, name, params} block:

```
- output:
  name:
  params:
```

Where, `output` represents the returned object, `name` identifies the function being executed, and `params` contains the function arguments. At runtime, `RunWorkflows()` parses these step definitions and orchestrates their execution sequentially, effectively reproducing the behavior of a piped R command.

- Raw → SDTM (workr + sdtm.oak): Raw eCRF data were transformed into SDTM using {workr}-driven workflows aligned with established `sdtm.oak` implementation patterns. The workflow mirrors the VS domain example from the official vignette, demonstrating how standardized SDTM mappings can be embedded within a reproducible YAML pipeline. The reference implementation is available here.
- SDTM → ADaM (workr + admiral): SDTM domains were reintroduced into the workflow engine and processed through admiral derivations to construct ADaM datasets. Functions such as `derive_vars_merged()` and `derive_param_map()` generated ADVS parameters and analysis values (e.g., MAP), with derivation logic expressed via !expr tags. This approach preserves transparency while ensuring transformations remain reproducible and audit-ready. The admiral vignette that demonstrates these functions can be found here.
- ADaM → TFLs (workr + gtsummary + safetyCharts): The resulting ADaM datasets fed directly into visualization steps, invoking `gtsummary` for tabular summaries and `safetyCharts` for graphical outputs. This enables generation of publication-ready TFLs within a fully traceable end-to-end pipeline, adaptable to an organization's visualization standards. Beyond static reporting, the same workflows can automatically refresh web-based or HTML applications as new data are derived, supporting near real-time exploratory analysis and safety review.
- ADaM → ARS (workr + cards): In parallel, the derived ADaM datasets were extended using cards to prototype ARS-aligned datasets and outputs. Although the ARS standard is still evolving, this example illustrates how emerging libraries can be integrated into the {workr} framework with minimal friction, enabling early adoption of new reporting standards within an existing, reproducible workflow.

Results / Conclusion

Each package preserved its modular design, while {workr} provided centralized workflow control, auditability, and reproducibility across the pipeline.

This pharmaverse case study demonstrates the feasibility of a fully open-source, end-to-end clinical reporting workflow. In this model, {workr} serves as the orchestration backbone, while pharmaverse packages deliver domain-specific transformations spanning SDTM, ADaM, TFLs, and ARS. Together, they form a modular, transparent, and automation-ready reporting architecture. We propose this integration as a blueprint for future collaboration between workflow frameworks and pharmaverse tools, advancing the industry toward interoperable, open-source clinical reporting standards.